COMPASS TEMPEST Sulfide

June 2026 Samples

2026-07-29

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed
sample_month = "June"
sample_year = "2026"
Run_Date = as.Date("06/16/2026", format = "%m/%d/%Y")
Run_by = "Zoe Read, Melanie Giessner"#Instrument user
Script_run_by ="Zoe Read"#Code user

Run_notes="All dups and related samples had values of 0.
11/16 spikes were good."#any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate\\d+?(:_202606:)?")
  stringr::str_subset("20260724", negate = TRUE)
(Raw_plates)
```

```
##  [1] "Raw Data/TEMPEST_Event2026_Plate1_20260616.csv"
##  [2] "Raw Data/TEMPEST_Event2026_Plate10_20260616.csv"
##  [3] "Raw Data/TEMPEST_Event2026_Plate11_20260617.csv"
##  [4] "Raw Data/TEMPEST_Event2026_Plate12_20260617.csv"
##  [5] "Raw Data/TEMPEST_Event2026_Plate13_20260617.csv"
##  [6] "Raw Data/TEMPEST_Event2026_Plate14_20260617.csv"
##  [7] "Raw Data/TEMPEST_Event2026_Plate15_20260617.csv"
##  [8] "Raw Data/TEMPEST_Event2026_Plate16_20260617.csv"
##  [9] "Raw Data/TEMPEST_Event2026_Plate2_20260616.csv"
## [10] "Raw Data/TEMPEST_Event2026_Plate3_20260616.csv"
## [11] "Raw Data/TEMPEST_Event2026_Plate4_20260616.csv"
## [12] "Raw Data/TEMPEST_Event2026_Plate5_20260616.csv"
## [13] "Raw Data/TEMPEST_Event2026_Plate6_20260616.csv"
## [14] "Raw Data/TEMPEST_Event2026_Plate7_20260616.csv"
## [15] "Raw Data/TEMPEST_Event2026_Plate8_20260616.csv"
## [16] "Raw Data/TEMPEST_Event2026_Plate9_20260616.csv"
```

```
#Sample ID Path
SampleID_path = "Raw Data/TEMPEST_S04_20260501-20260612_corrected.csv"
```

```r
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606_processed.csv"
rerun_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606_reruns.csv"
```

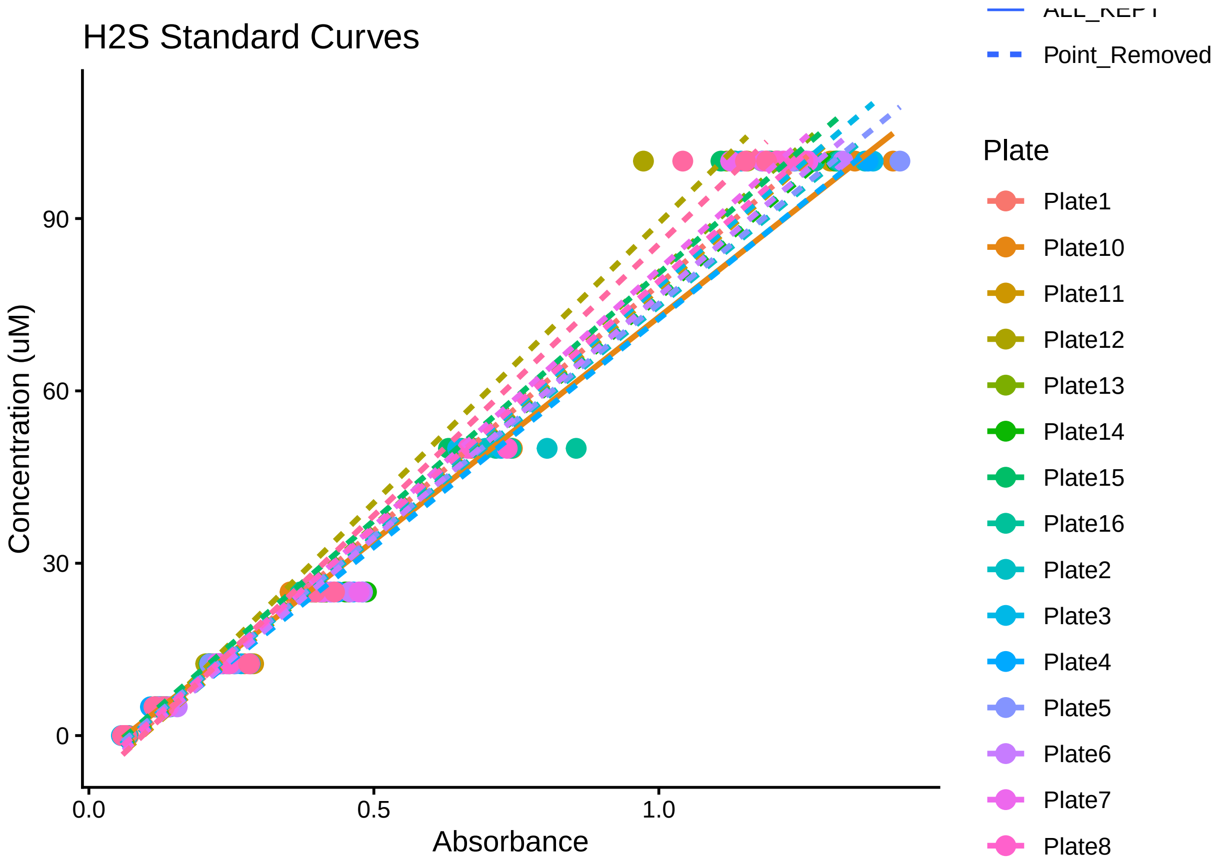
##Read in data

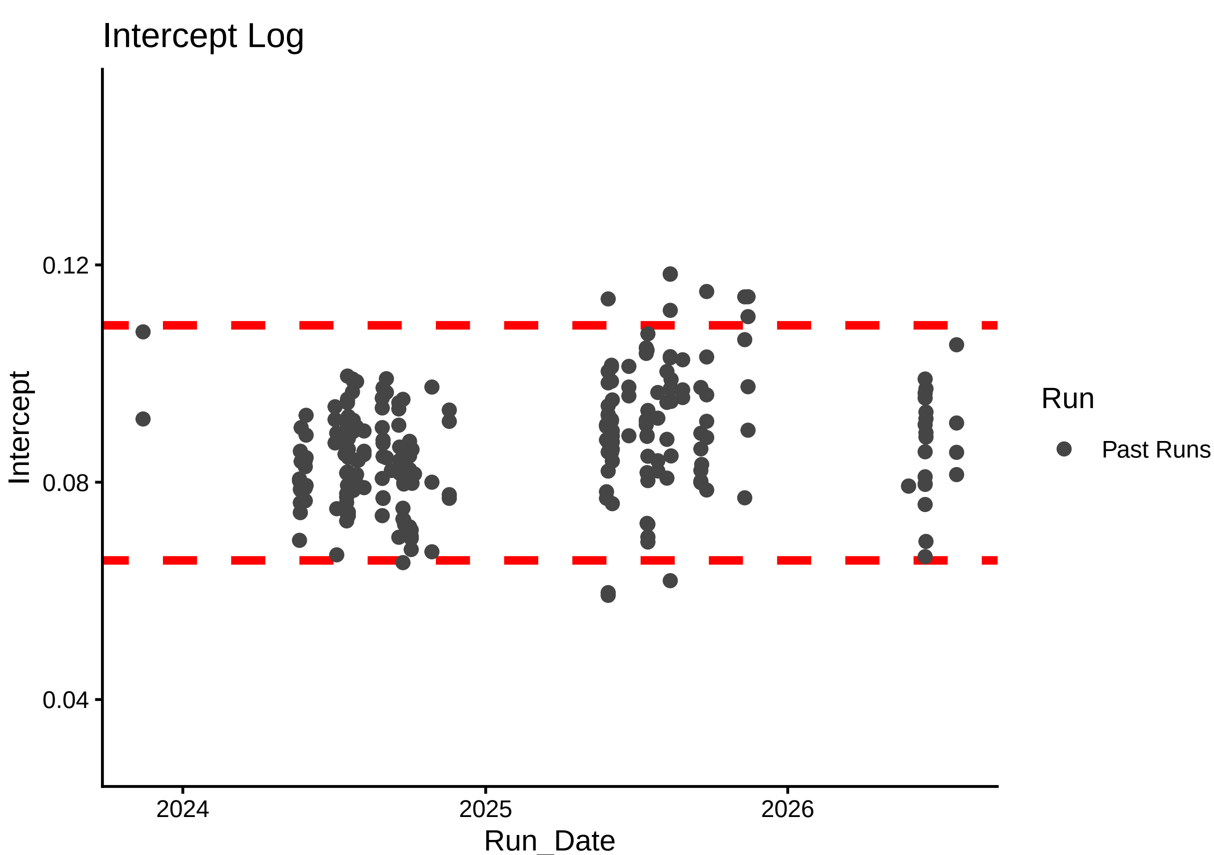
##Remove High CV Stds

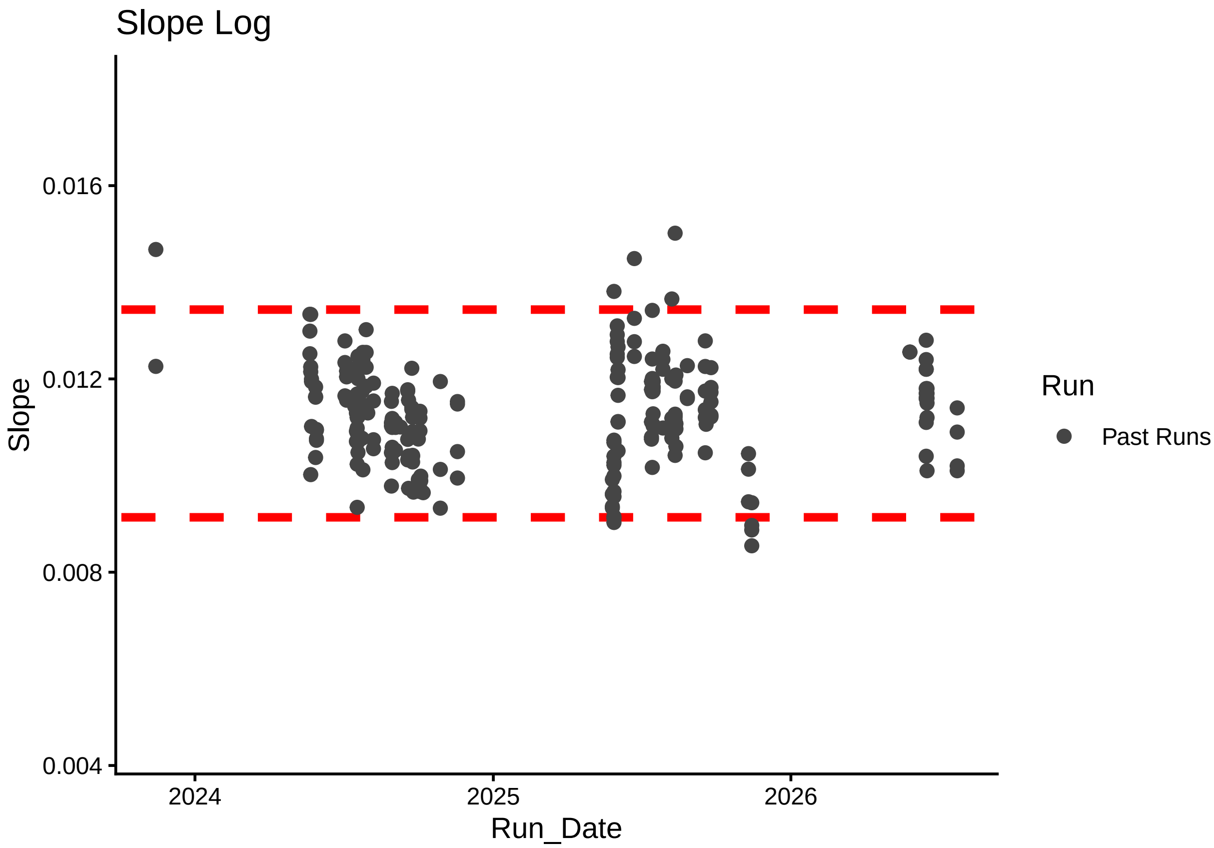
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

Plot Standard Curves







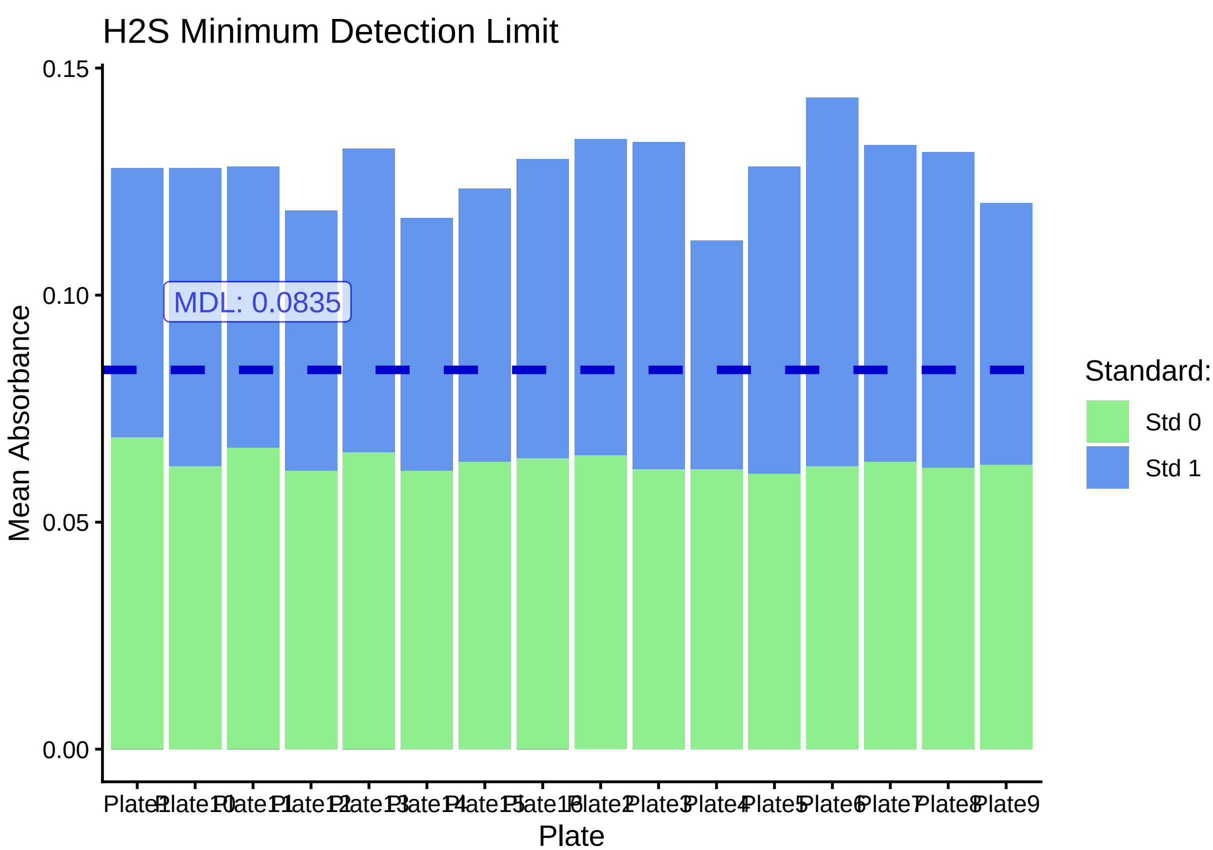
Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate10	R2_Good	CV_Good
Plate11	R2_Good	CV_Good
Plate12	R2_Bad	CV_Good
Plate13	R2_Good	CV_Good
Plate14	R2_Bad	CV_Good
Plate14	R2_Bad	CV_High
Plate15	R2_Good	CV_Good
Plate15	R2_Good	CV_High
Plate16	R2_Bad	CV_Good
Plate2	R2_Good	CV_Good
Plate3	R2_Good	CV_Good
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good
Plate6	R2_Good	CV_Good
Plate6	R2_Good	CV_High
Plate7	R2_Good	CV_Good
Plate8	R2_Good	CV_Good
Plate9	R2_Bad	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate10	KEPT
Plate11	KEPT

Plate	Curve_RMVD
Plate12	REMOVED
Plate13	KEPT
Plate14	REMOVED
Plate15	KEPT
Plate16	REMOVED
Plate2	KEPT
Plate3	KEPT
Plate4	KEPT
Plate5	KEPT
Plate6	KEPT
Plate7	KEPT
Plate8	KEPT
Plate9	REMOVED

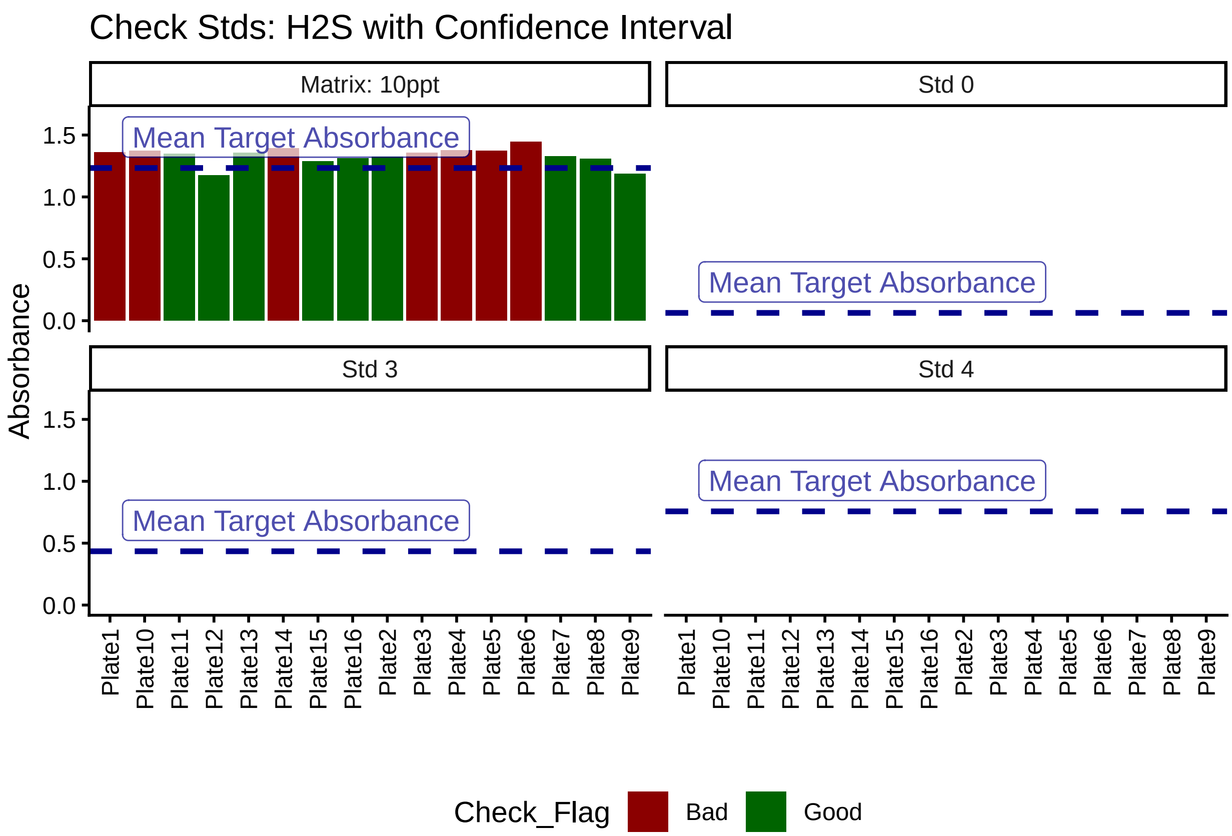
Method Detection Limit



Compare Check Standards to Standards

```
## Warning in min(x): no non-missing arguments to min; returning Inf

## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in min(x): no non-missing arguments to min; returning Inf

## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in min(x): no non-missing arguments to min; returning Inf
```

```
## Warning in max(x): no non-missing arguments to max; returning -Inf

## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
```



Display Any Check Flags

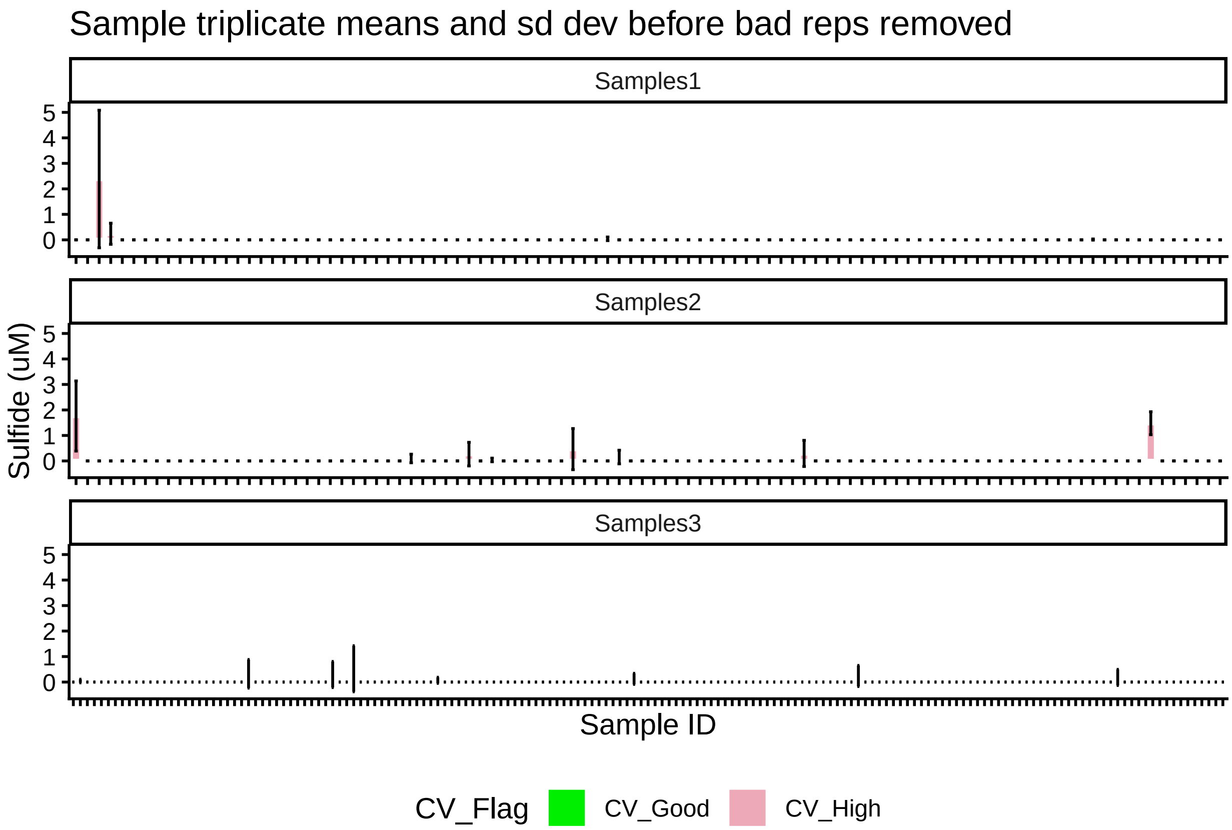
Plate	CHKs_Flag
Plate1	MC: 10ppt S5 Chk_Bad
Plate10	MC: 10ppt S5 Chk_Bad
Plate14	MC: 10ppt S5 Chk_Bad
Plate3	MC: 10ppt S5 Chk_Bad
Plate4	MC: 10ppt S5 Chk_Bad
Plate5	MC: 10ppt S5 Chk_Bad
Plate6	MC: 10ppt S5 Chk_Bad

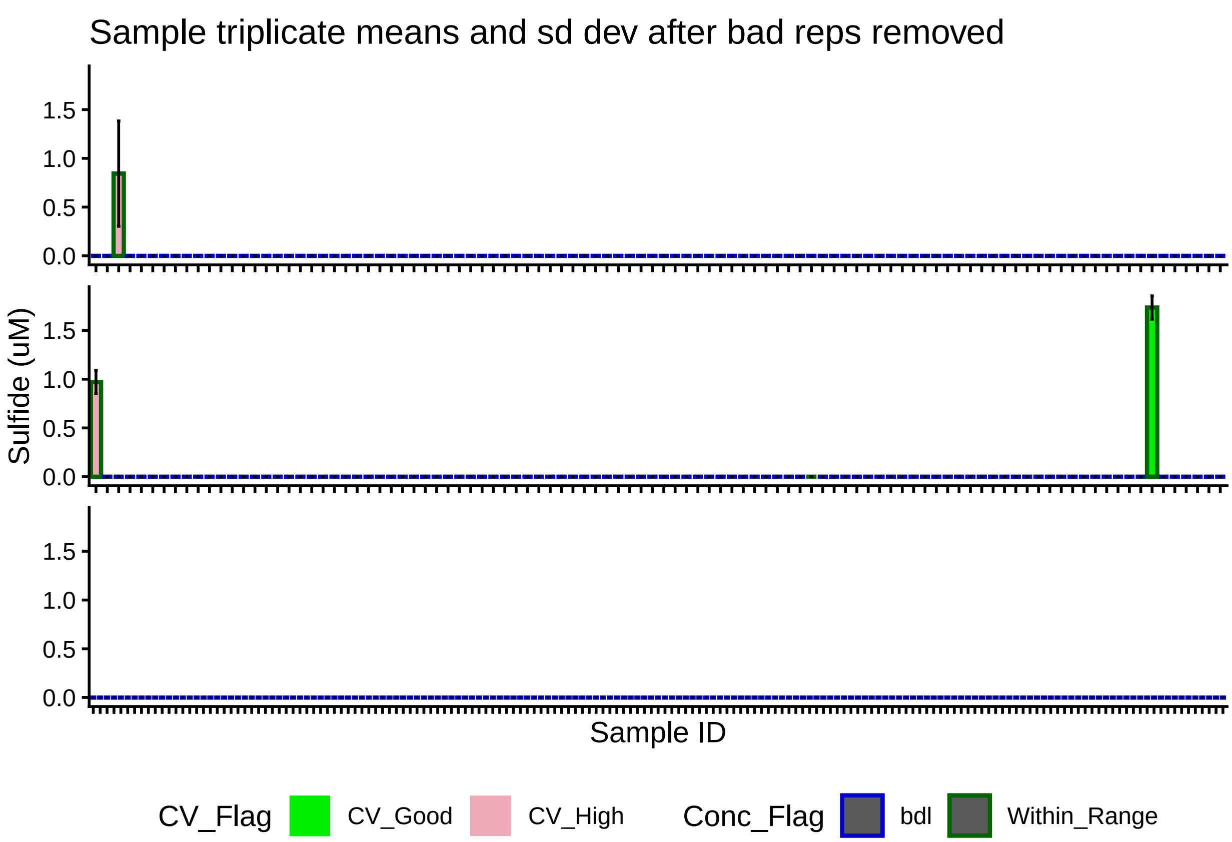
Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

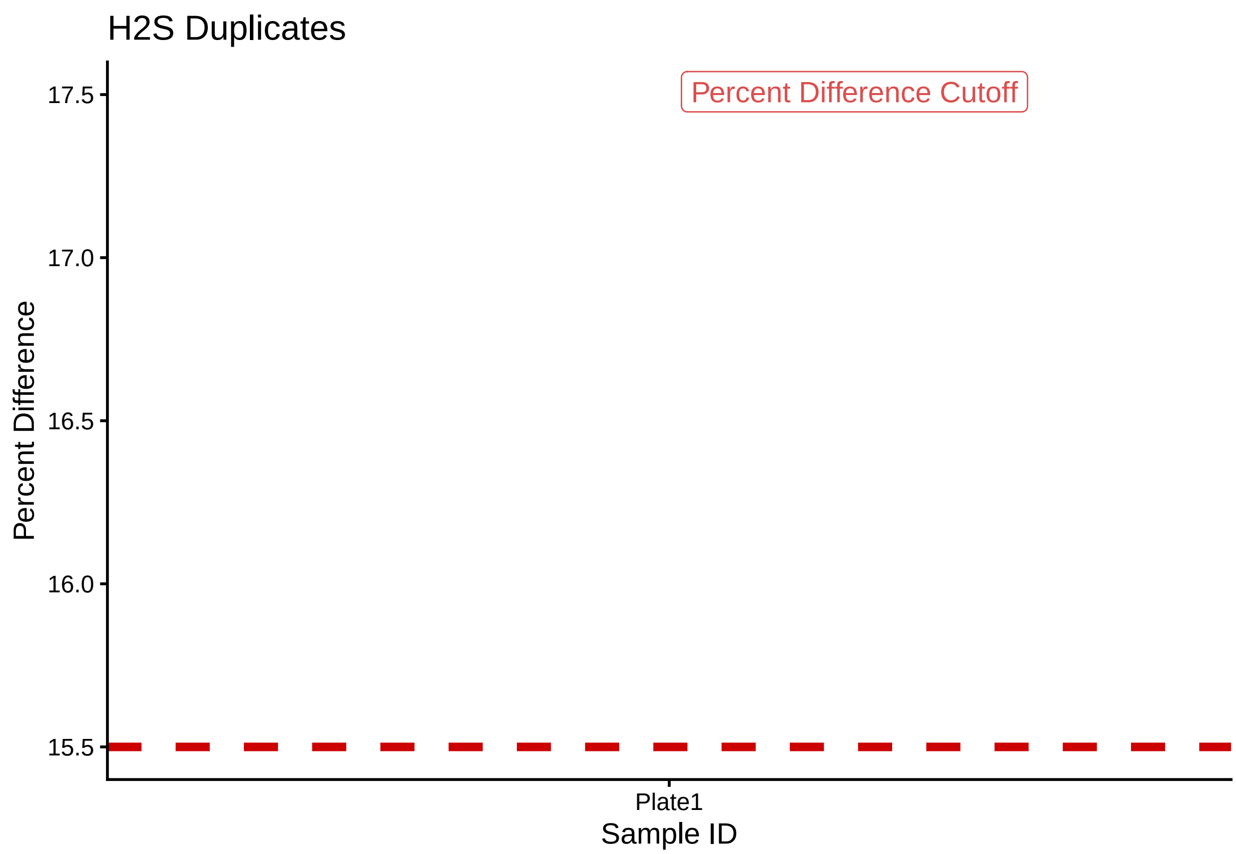
Plot Samples





Duplicate QAQC

```
## Warning: No shared levels found between `names(values)` of the manual scale and the
## data's fill values.
```

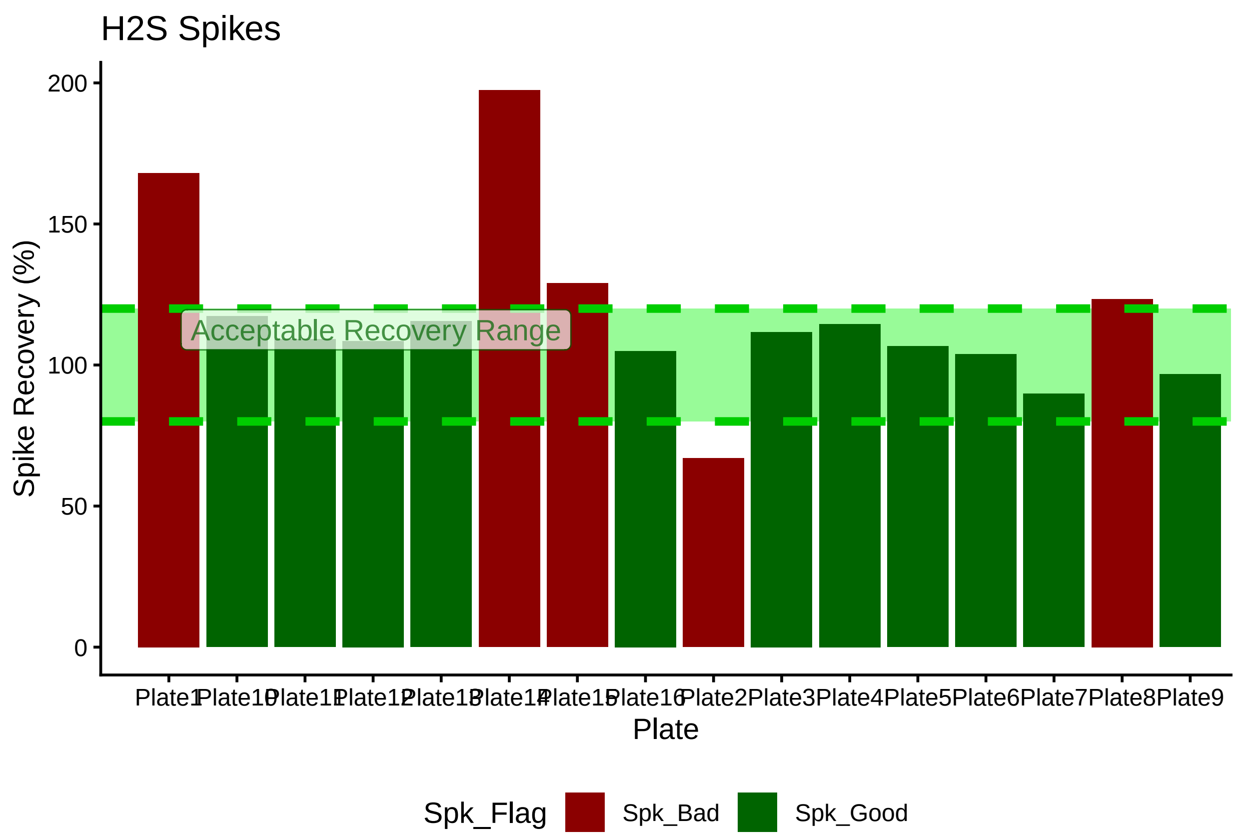


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate10	Dup_Good
Plate11	Dup_Good
Plate12	Dup_Good
Plate13	Dup_Good
Plate14	Dup_Good
Plate15	Dup_Good
Plate16	Dup_Good
Plate2	Dup_Good
Plate3	Dup_Good
Plate4	Dup_Good
Plate5	Dup_Good
Plate6	Dup_Good
Plate7	Dup_Good
Plate8	Dup_Good
Plate9	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate10	Spk_Good
Plate11	Spk_Good
Plate12	Spk_Good
Plate13	Spk_Good
Plate14	Spk_Bad
Plate15	Spk_Bad
Plate16	Spk_Good
Plate2	Spk_Bad
Plate3	Spk_Good
Plate4	Spk_Good
Plate5	Spk_Good
Plate6	Spk_Good
Plate7	Spk_Good
Plate8	Spk_Bad
Plate9	Spk_Good

```
## [1] "Spikes Pass"
```

Check for reruns and replace

##Add Flags

##Remove Synoptic samples and rename TEMPEST samples

```
df_all_clean <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "SWH|GCW|GWI|MSM")) %>%
  filter(Sample_ID != "141") #%>%
  # mutate(Sample_ID = gsub("TMP_2026_EVENT_141A", "141A", Sample_ID)) %>%
```

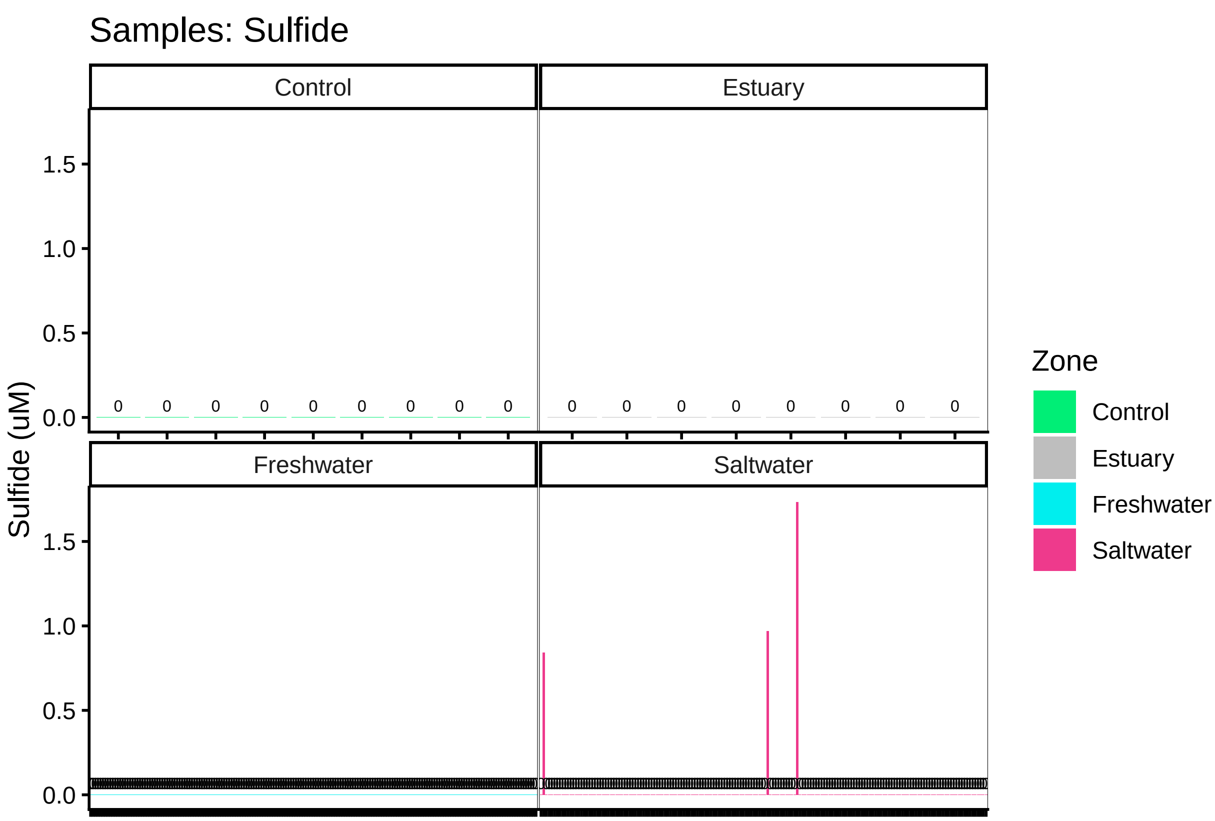
##Merge with Sample ID list to get actual sample IDs

##Pull in active porewater inventory

##Check samples against metadata

```
## ***All sample IDs are present in metadata.***
```

Visualize Data by Plot



##Export Data

END